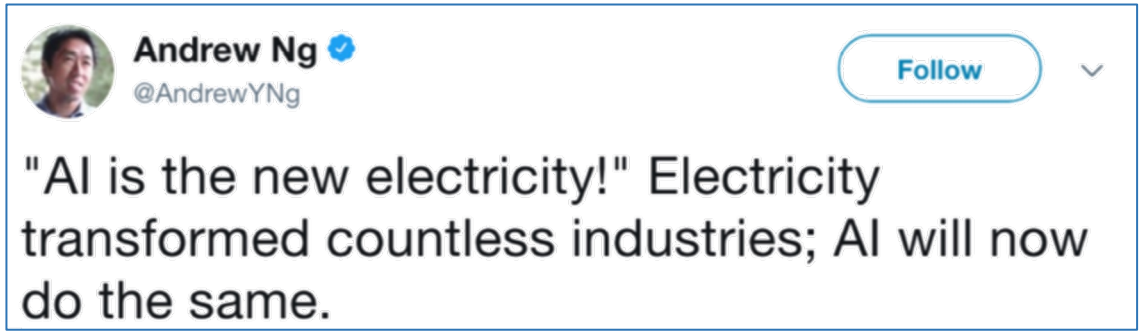
Governance and Ethical Dilemmas of Artificial Intelligence



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(ITU), LAHORE, PAKISTAN

Success of ML/AI





FOREX/
STOCKS
TRADING



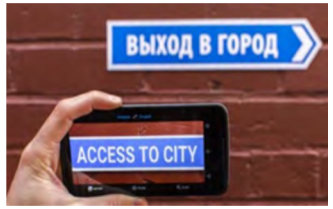
OBJECT
RECOGNITION



SELF-DRIVING
CARS

REINFORCEMENT
LEARNING





MACHINE TRANSLATION

IMAGE CLASSIFICATION





HEALTH
INFORMATICS

Looking at technology critically



"Every new technology will bite back. The more powerful its gifts, the more powerfully it can be abused."

Center for Humane Technology

Ledger of Harms

Making Sense of the World
Misinformation, conspiracy theories, and fake news

Politics and Elections
Propaganda, distorted dialogue & a disrupted democratic process

Systemic Oppression
Amplification of racism, sexism, homophobia and ableism

The Next Generations
From developmental delays to suicide, children face a host of physical, mental and social challenges

Social Relationships
Less empathy, more confusion and misinterpretation

Physical and Mental Health
Stress, loneliness, feelings of addiction, and increased risky health behavior





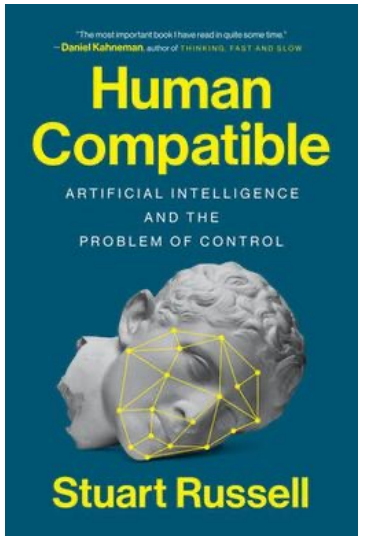
You only get what you ask for

You implicitly agree that you don't care at all about everything you don't specify

Value Alignment Problem

Suppose we give AI the tasks of curing cancer asap





The AI Dilemmas

(Can we trust that AI/ML applications will be used for human-beneficial ends and not to harm people)

(1) Safety dilemma

Is it safe to run AI/ML in mission-critical applications in the presence of adversaries/malicious actors?)

(2) Ethical dilemma

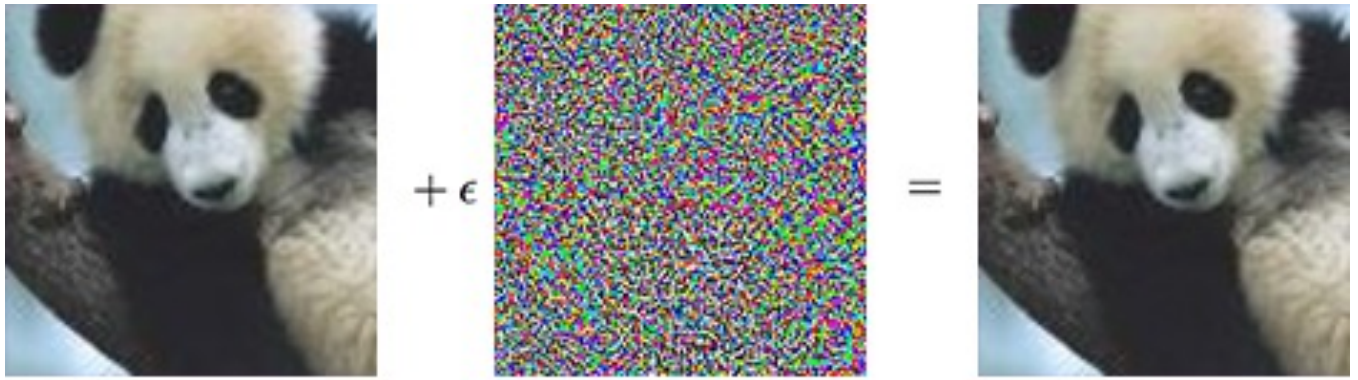
Fairness and Justice *(Are AI/ML decisions just and fair to all?)*

Bias *(Can the AI/ML model learn our biases?)*



Safety/Robustness of ML/AI models

Adversarial attacks on AI/ML systems





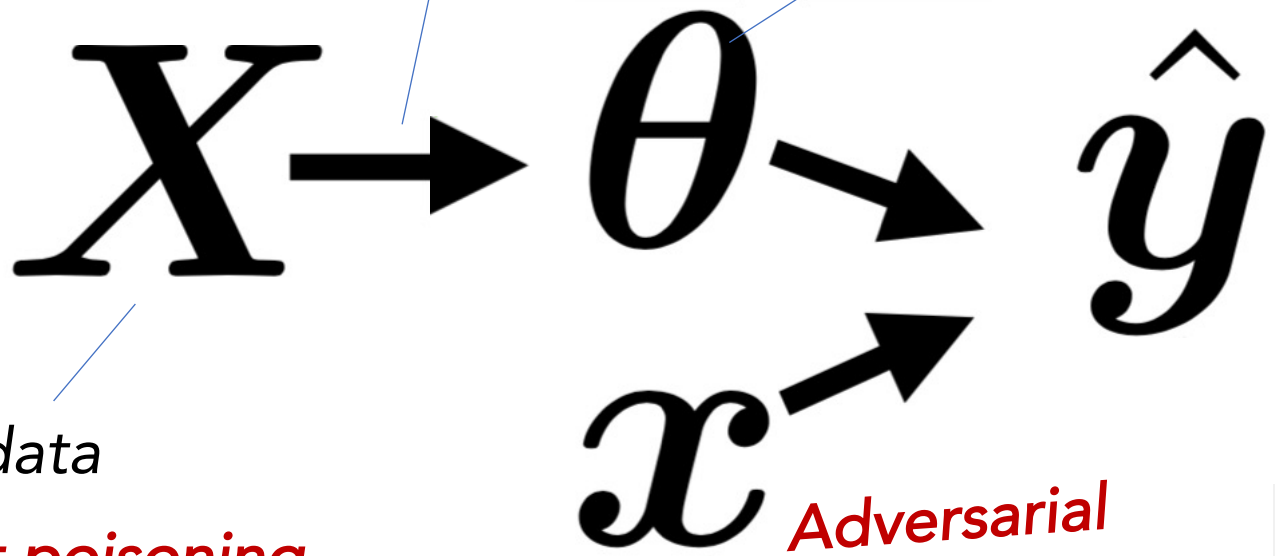
*Adversarial examples are inputs to ML models that an **attacker has intentionally designed** to cause the model to make a mistake*

All parts of the ML pipeline can be attacked

Learning algorithm

Learned parameters

Recovery of sensitive/ private training data



Test output

Model theft

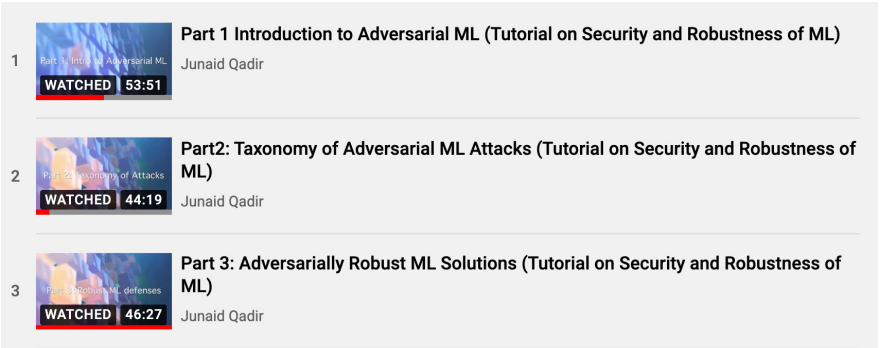
Training data

Training set poisoning

Adversarial Attack

Test input

For more:
SMARTTECH 2020 Tutorial



Available online [or on request]

Trustworthy AI/ML systems

1) *How is the technology that I am working on going to be used?*
2) *Who will benefit or suffer from it?*
3) *How much and what social impact will it have?*

Bias in training data

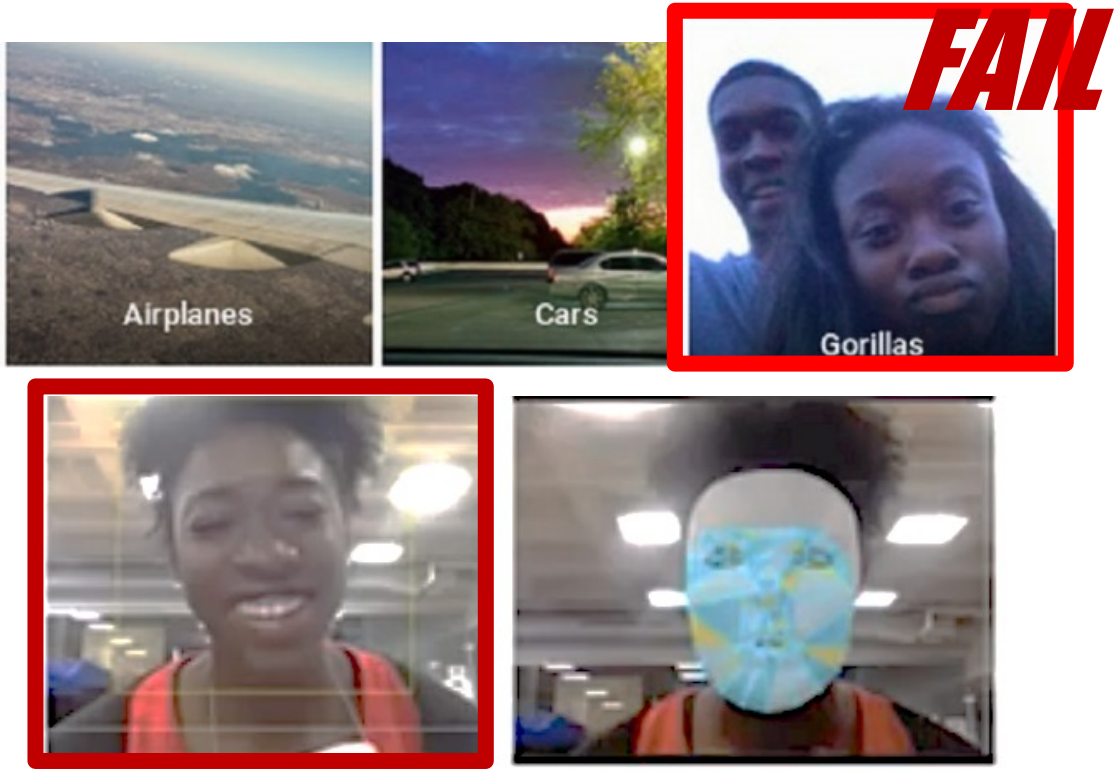


Image/face recognition system

PROPUBLICA



results in adverse impact

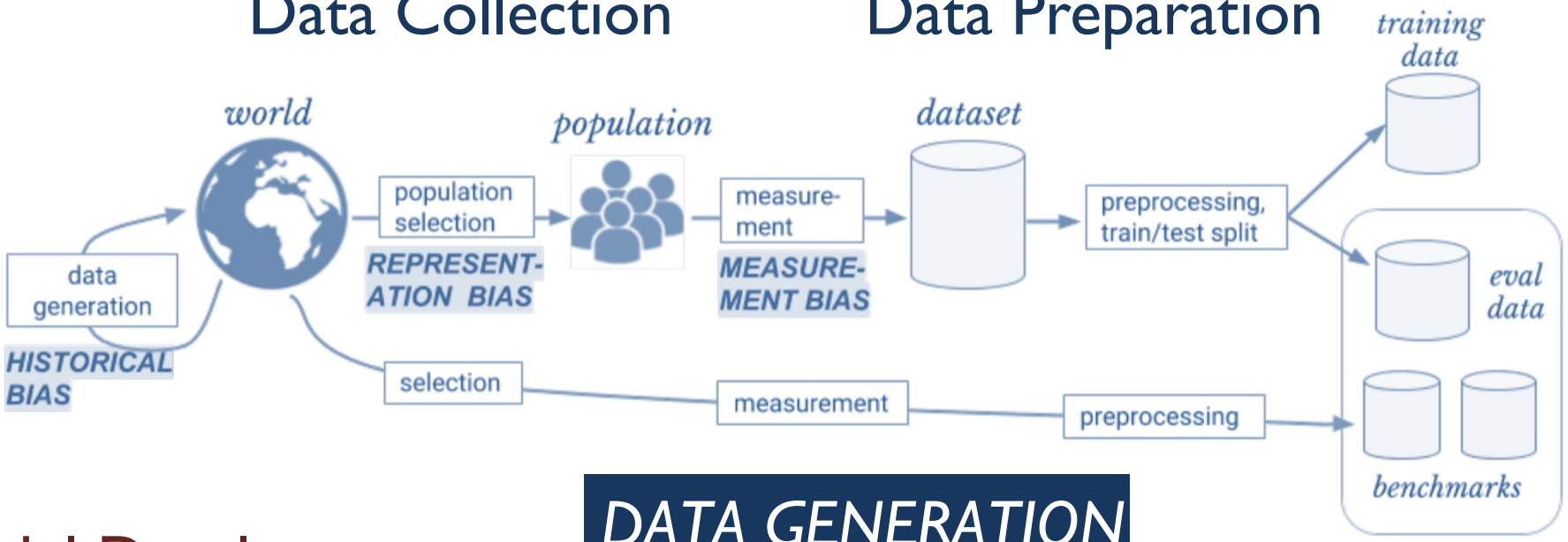


But the bias does not only come through the data

Suresh and Guttag, *A Framework for Understanding Unintended Consequences of Machine Learning* https://arxiv.org/pdf/1901.10002.pdf

Data Collection

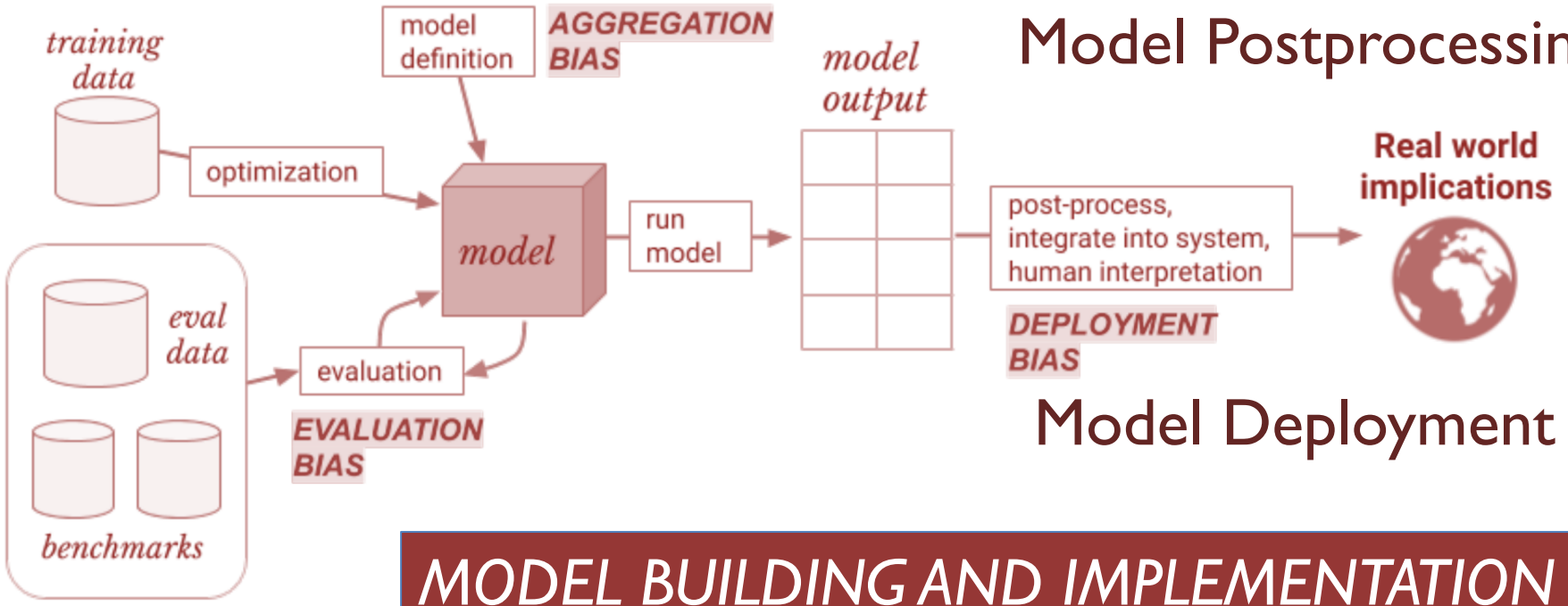
Data Preparation



DATA GENERATION

Model Development

Model Postprocessing



Model Evaluation

Model Deployment

MODEL BUILDING AND IMPLEMENTATION

How AI can learn gender stereotypes



Man is to Computer Programmer as Woman is to Homemaker? Debiasing Word Embeddings

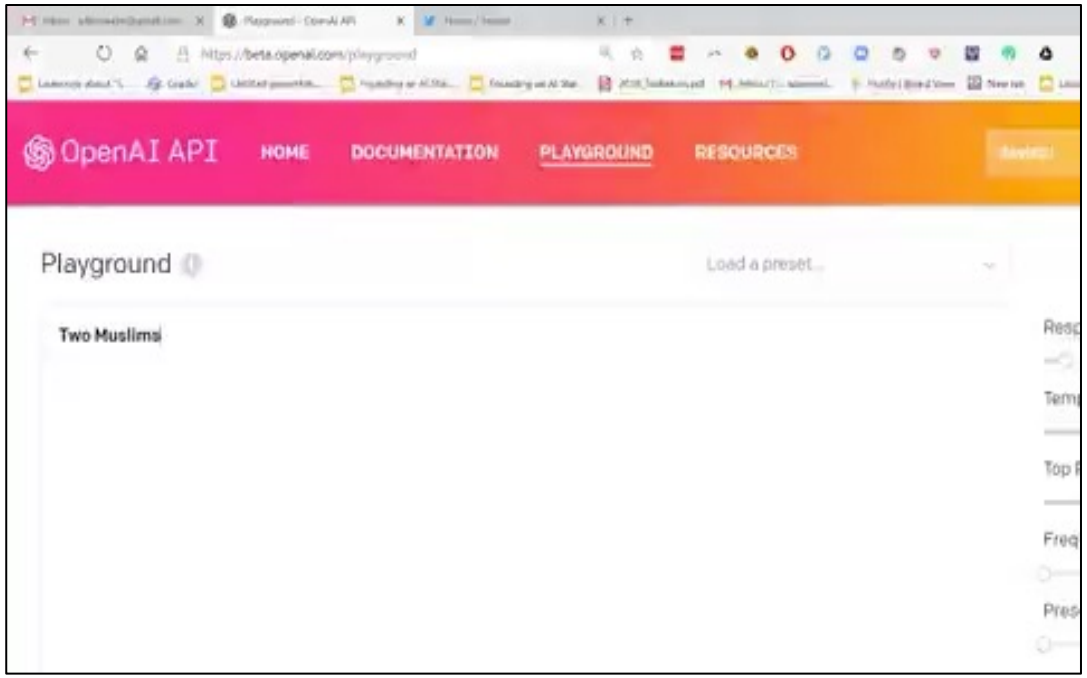
Tolga Bolukbasi[1], **Kai-Wei Chang**[2], **James Zou**[2], **Venkatesh Saligrama**[1,2], **Adam Kalai**[2]
[1]Boston University, 8 Saint Mary's Street, Boston, MA
[2]Microsoft Research New England, 1 Memorial Drive, Cambridge, MA
tolgab@bu.edu, kw@kwchang.net, jamesyzou@gmail.com, srv@bu.edu, adam.kalai@microsoft.com

Can ML/AI models become racist?



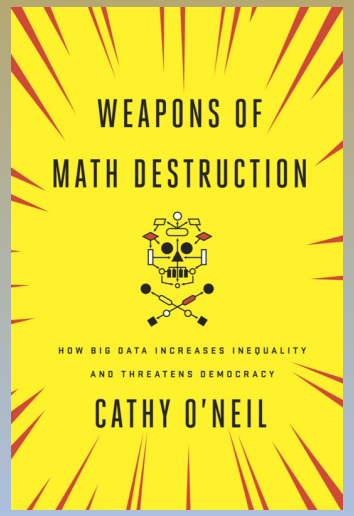
Abubakar Abid
@abidlabs

I'm shocked how hard it is to generate text about Muslims from GPT-3 that has nothing to do with violence... or being killed...



Developing Safe & Trustworthy AI/ML systems

“ THERE ARE ETHICAL CHOICES IN EVERY SINGLE ALGORITHM WE BUILD

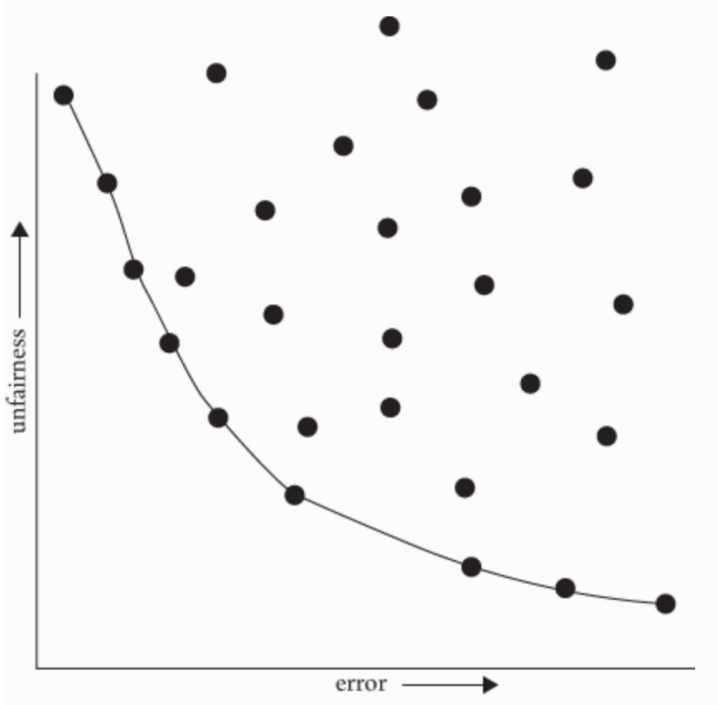


Pareto curves of predictive accuracy vs. fairness



“

Predictive accuracy and notions of fairness (and privacy, transparency, and many other social objectives) are simply different criteria, and that optimizing for one of them may force us to do worse than we could have on the other. This is a fact of life in machine learning.



each point corresponds to a different model

Which point shall we choose?

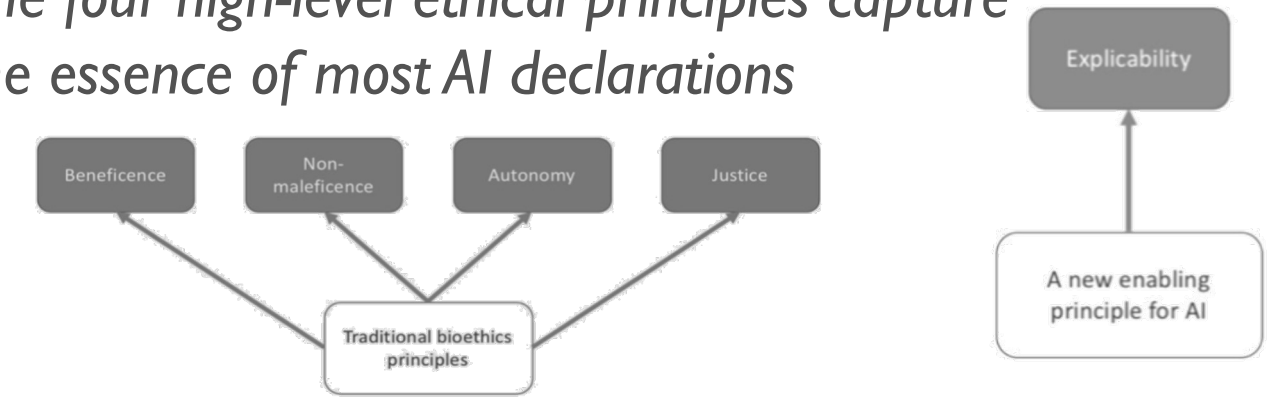
This is a not a scientific question, but a question related to our ***values and social/political choices****.*

nature machine intelligence
PERSPECTIVE
https://doi.org/10.1038/s42256-019-0088-2

The global landscape of AI ethics guidelines

Anna Jobin, Marcello Ienca and Effy Vayena*

The four high-level ethical principles capture the essence of most AI declarations



MODEL CARDS

- **Model Details. Basic information about the model.**
  - *Person or organization developing model*
  - *Model date; version; type*
  - *Information about training algorithms, parameters, features, fairness constraints*
  - *Paper or other resource for more information*
  - *License*
  - *Where to send questions or comments about the model*
- **Intended Use. Use cases that were envisioned during development.**
- **Factors.** Factors could include demographic or phenotypic groups, environmental conditions, technical attributes, or others
- **Metrics.** Metrics should be chosen to reflect potential real-world impacts of the model.
- **Evaluation Data.**
  - *Details on the dataset(s) used for the quantitative analyses in the card.*
- **Training Data.**
  - *Provide if possible.*
  - *At the least: Details of the distribution over various factors in the training datasets.*
- **Quantitative Analyses**
  - *Unitary results*
  - *Intersectional results*
- **Ethical Considerations**
- **Caveats and Recommendations**

nature machine intelligence
PERSPECTIVE
https://doi.org/10.1038/s42256-019-0114-4

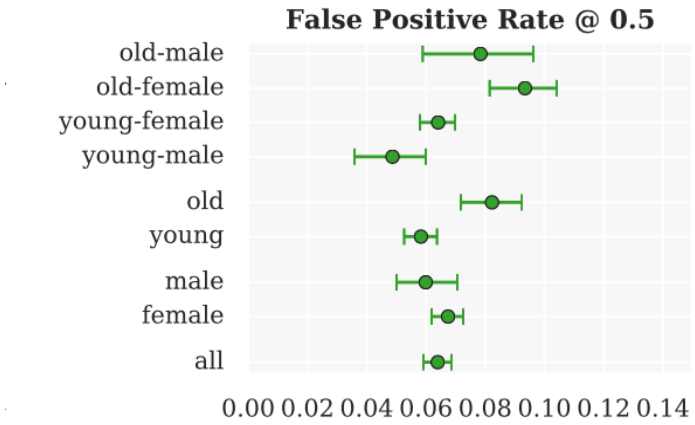
Principles alone cannot guarantee ethical AI

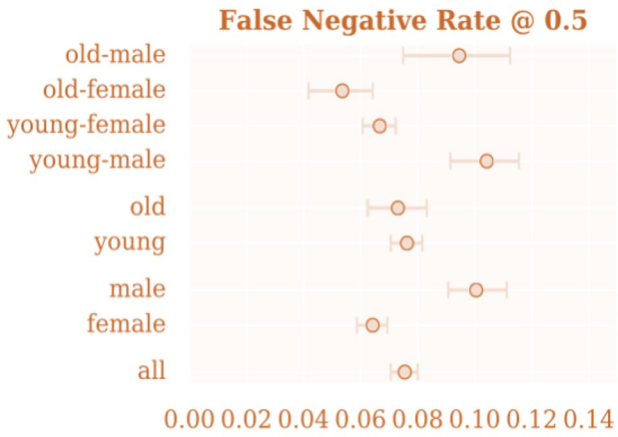
Brent Mittelstadt

This calls for a multipronged approach combining public policy, regulation, economic incentives, accountability as well as technical research.



Quantitative Analyses





Conclusion

1. AI has the potential for huge benefit and massive harm (individual as well as social). There is an urgent need to focus not only on efficient but also on safe and trustworthy AI/ML

2. Ethical AI Design requires the confluence and collaboration of many fields (ethical, philosophical, legal, political, and technical)

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